

AETHERIX

Autonomous Extraterrestrial High-throughput Enhancing Routing
and Inter-planetary eXchange

Building an Interplanetary Communication Network with DTN,
Quantum Communication, and Space-Based Infrastructure for Mars Mission Support

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matx104.github.io/AETHERIX | github.com/matx104/AETHERIX

A decorative graphic consisting of a dotted arc spanning the width of the page. It features two colored dots: a teal dot on the left and an orange dot on the right, connected by a series of small dots.

PRESENTATION AGENDA

01 The Challenge

Why interplanetary communication is hard

02 AETHERIX Architecture

DTN + AI Routing + Quantum Security

03 DTN & Bundle Protocol v7

Store-and-forward networking foundation

04 5-Tier Network Topology

241 nodes across Earth and Mars

05 Optical Link Budget

1550 nm laser performance analysis

06 RL-Based Routing

Multi-agent federated Q-learning

07 Quantum Security (QKD)

BB84/E91 with repeater chains

08 Orbital Mechanics

Contact windows and synodic period

09 Mars Mission Scenario

End-to-end simulation walkthrough

10 Performance Comparison

AETHERIX vs current systems

11 Roadmap & Standards

CCSDS, IETF, deployment phases

12 Conclusion & Q&A

Summary and live demo

WHAT IS AETHERIX?

Autonomous Extraterrestrial High-throughput Enhancing Routing and Inter-planetary eXchange

What is AETHERIX?

An architecture for delay-tolerant networking (DTN) between Earth and Mars, combining:

- Bundle Protocol v7 (BPv7) via ION-DTN
- Reinforcement Learning routing (replacing static CGR)
- Quantum Key Distribution (QKD) for security
- Hybrid optical/RF communications
- 5-tier topology with 241 nodes

The Problem

- Earth-Mars: 54.6M – 401M km distance [3]
- One-way light delay: 3 – 22 minutes [3]
- Solar conjunction: 2-week total blackout
- TCP/IP fundamentally cannot work [12]
- Current RF: 0.5 – 6 Mbps (NASA MRO) [1]

AETHERIX delivers a complete interplanetary networking solution

combining DTN protocols, AI-driven routing, quantum-secure keys, and hybrid optical/RF links for Mars mission support.

10-100×

Target

>95%

Availability (target)

241

Nodes [A2]

480

Tests

THE DISTANCE

Why Space Communication is Fundamentally Different

Earth → Mars	Distance	Light Time	Data Rate
Perihelion (closest)	54.6M km	~3 min	100-200 Mbps
Average	225M km	~12.5 min	10-20 Mbps
Aphelion (farthest)	401M km	~22 min	2-5 Mbps

TCP/IP Assumption	Space Reality	Impact
RTT < 1 second	6 – 44 min RTT [3]	Timeout failure
Always connected	Scheduled contacts	Connection drops
End-to-end path	No persistent path	Routing impossible
Fast acknowledgments	ACKs take minutes [3]	Window collapse
Low packet loss	High loss (optical)	Retransmit sto

"At aphelion, a single ping-pong takes 44 minutes.
No TCP session can survive that. We need an entirely
different networking paradigm."

THE ANSWER: DELAY-TOLERANT NETWORKING

Decouple Communication from Connectivity



Key Insight: DTN moves reliability from end-to-end to hop-by-hop. Each custodian is responsible until the next node accepts custody — enabling communication even with 44-minute round-trip delays.

SYSTEM ARCHITECTURE

5 Integrated Modules for Interplanetary Communication

Infrastructure

Link budget calculations, optical/RF analysis

Routing

RL agents, BPv7 bundles, forwarding engine

Security

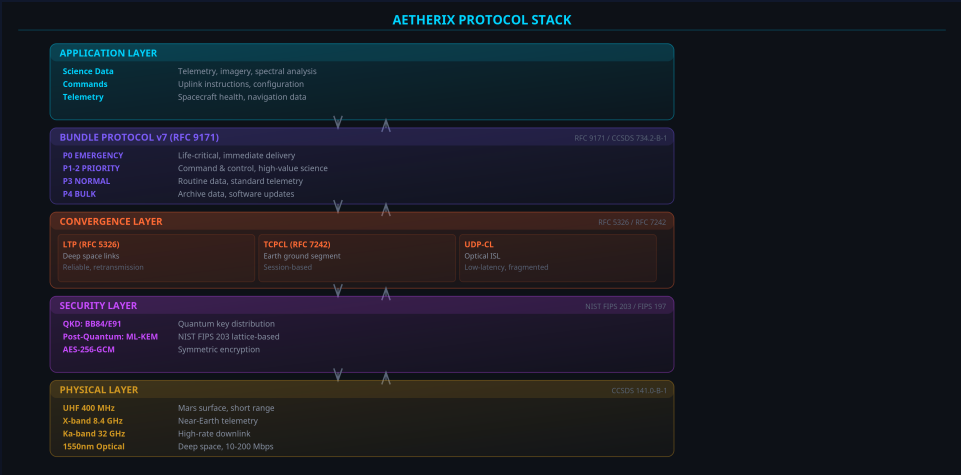
BB84/E91 QKD, repeater chains, privacy amplification

Orbital

Contact windows, Doppler, celestial body database

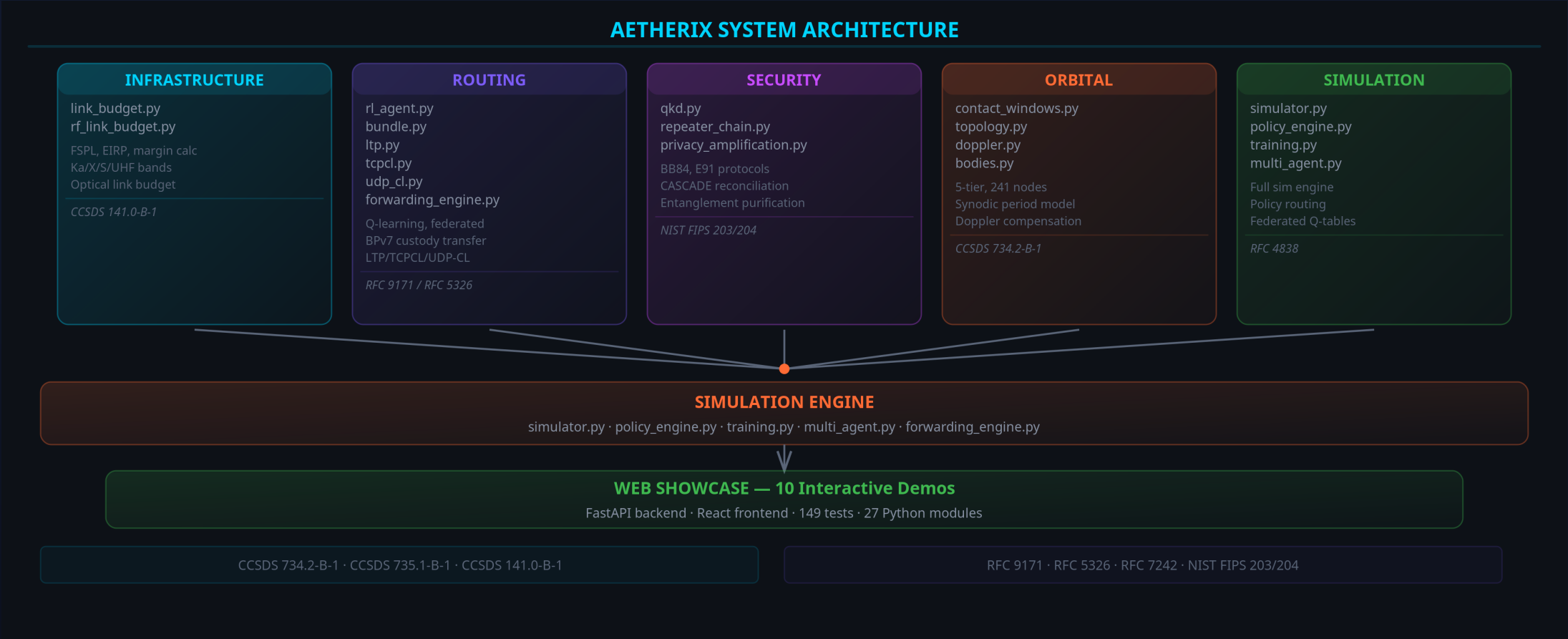
Simulation

End-to-end scenario engine, policy routing



Layer	Module	Engine	Showcase
Infrastructure	link_budget.py	OpticalLinkBudget	Earth-Mars scenarios
Routing	rl_agent.py	RLRoutingAgent	Federated Q-learning
Security	qkd.py	BB84/E91	Quantum key exchange
Orbital	contact_windows.py	ContactPredictor	Synodic period
Simulation	simulator.py	SimEngine	Full mission sim

SYSTEM ARCHITECTURE DIAGRAM



BUNDLE PROTOCOL v7 — DEEP DIVE

RFC 9171 — The Foundation of Interplanetary Networking

APPLICATION LAYER

Science data, commands, imagery

BUNDLE PROTOCOL v7 (RFC 9171)

Store-and-Forward | Custody Transfer | Priority (P0-P4)

CONVERGENCE LAYERS

LTP (deep space) | TCPCL (Earth) | UDP-CL (optical ISL)

PHYSICAL LAYER

Optical 1550nm (2-200 Mbps) | RF Ka-band | UHF (surface)

HOW IT WORKS

1. Source creates bundle (payload + metadata + lifetime)
2. Forward to next hop when link becomes available
3. Store locally during outages and disruptions
4. Custody transfer — next node accepts responsibility
5. Repeat hop-by-hop until destination reached

STANDARDS COMPLIANCE

RFC 9171 — Bundle Protocol Version 7 [9]

RFC 4838 — DTN Architecture [12]

RFC 5326 — Licklider Transmission Protocol [10]

RFC 9172 — Bundle Protocol Security (BPSec)

CCSDS 734.2-B-1 — CCSDS Bundle Protocol [2]

CCSDS 734.3-B-1 — Schedule-Aware Bundle Routing

Priority	Name	Delay Target	Example
P0	Emergency	< 1 min	Anomaly alerts
P1	Expedited	< 30 min	Solar storm warnings
P2	Standard	< 24 hrs	Science data
P3	Normal	< 7 days	Telemetry
P4	Bulk	< 30 days	Software updates

5-TIER NETWORK TOPOLOGY

241 Nodes — No Single Point of Failure

TIER 1: Earth Ground (6 nodes)

DSN Goldstone, Madrid, Canberra + Mission Control

TIER 2: Earth Orbital (51 nodes)

3 GEO relays + 48 LEO laser constellation

TIER 3: Deep Space (4 nodes)

ES-L4, ES-L5 Lagrange relays + Transfer orbit

TIER 4: Mars Orbital (4 nodes)

2 Areostationary + 2 polar orbiters

TIER 5: Mars Surface (176 nodes)

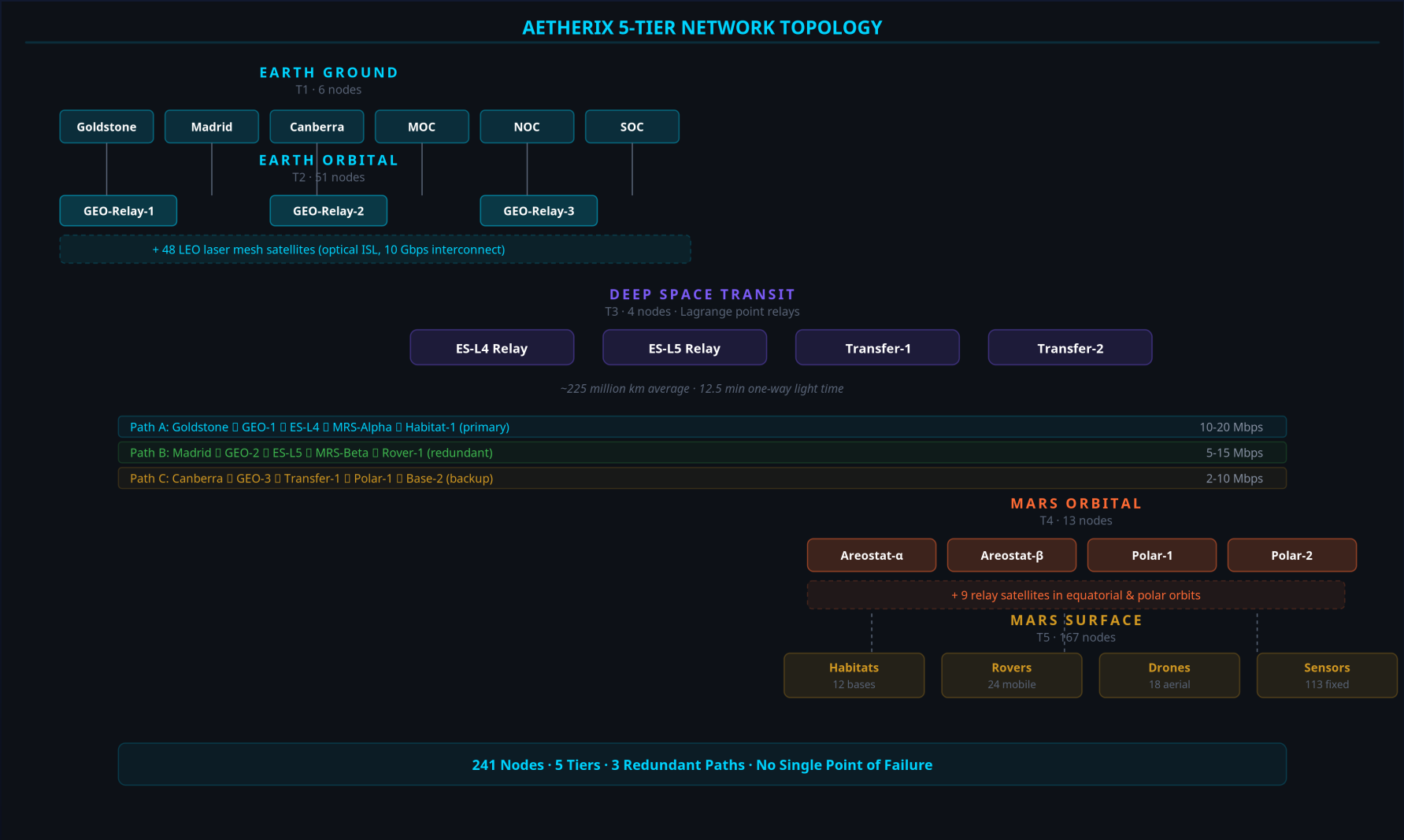
Bases, rovers, drones, sensor network

DESIGN PHILOSOPHY

Multiple redundant paths at every tier ensure no single point of failure. Lagrange point relays provide coverage during solar conjunction blackouts. Quantum repeaters co-located at L4/L5 enable end-to-end key distribution.

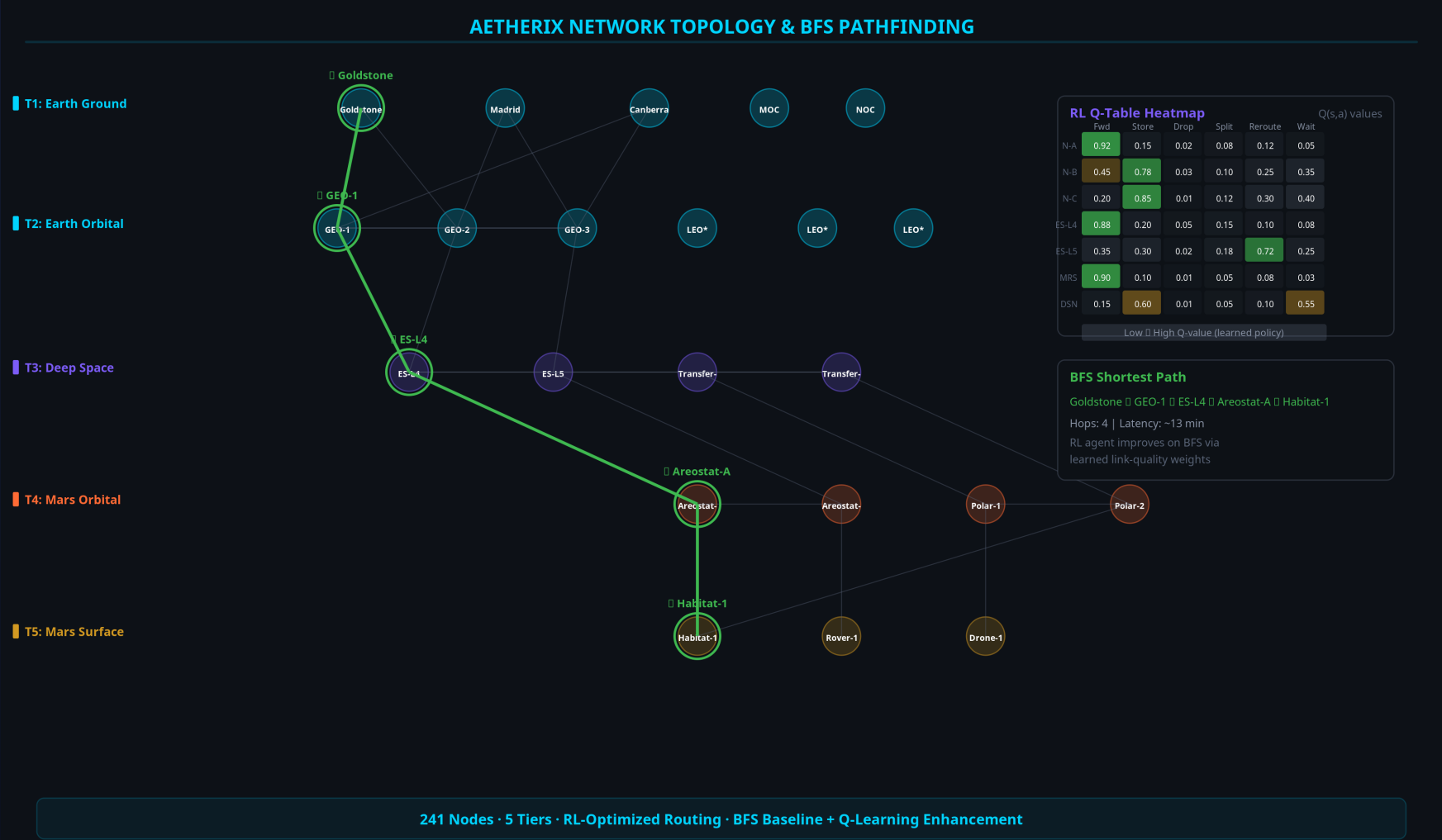
5-TIER NETWORK DIAGRAM

241 Nodes from Earth to Mars Surface



NETWORK TOPOLOGY GRAPH

241 Nodes – BFS Pathfinding + RL-Optimized Routing



OPTICAL COMMUNICATIONS

1550 nm Laser — 10–100× Capability Over RF

200 Mbps

Peak Data Rate

At perihelion (54.6M km)

1550 nm

Wavelength

Telecom heritage lasers

33×

vs Current RF

MRO peak: 6 Mbps

KEY EQUATIONS (CCSDS 141.0-B-1 [5])

$FSPL = (4\pi d/\lambda)^2$ Free-Space Path Loss [5]

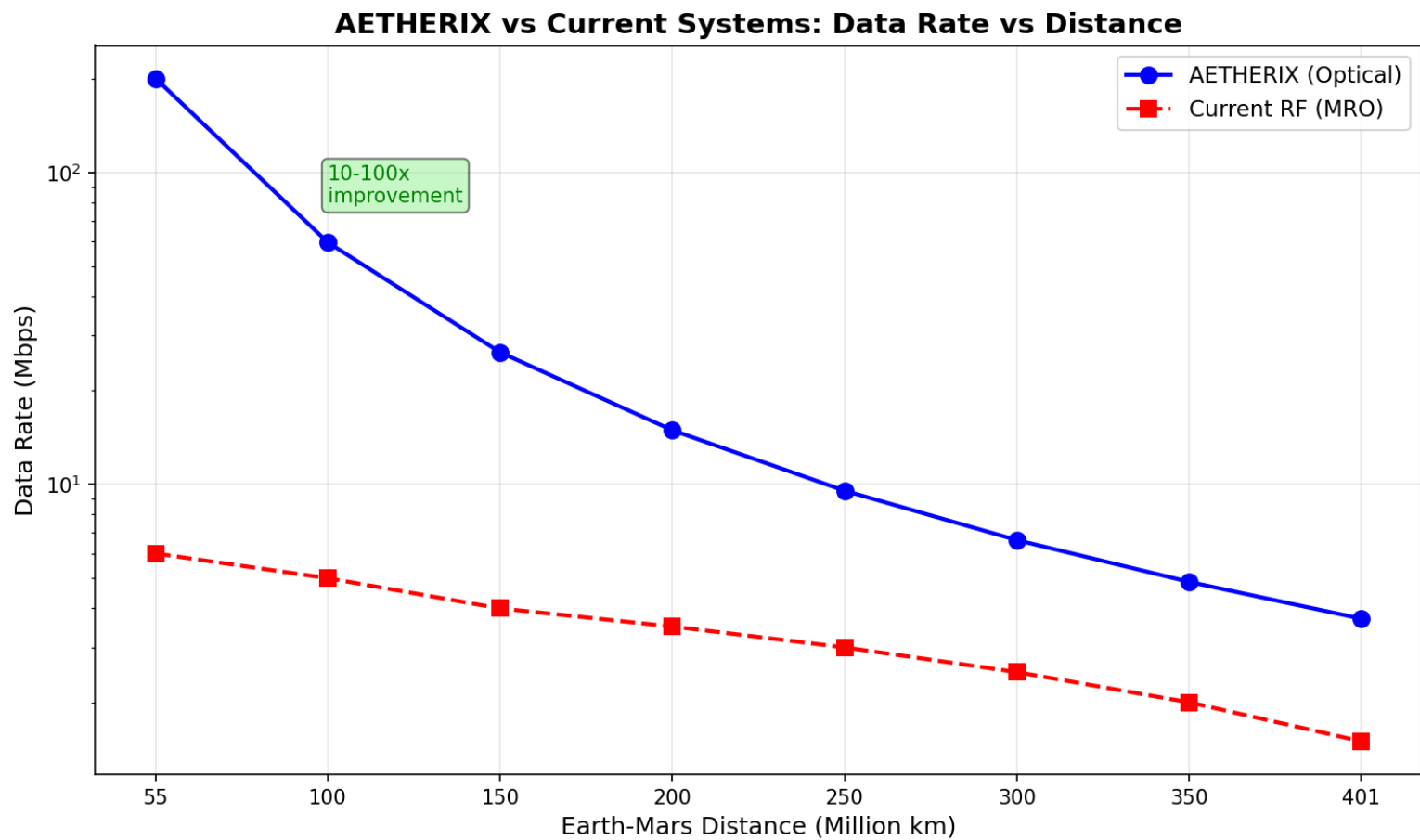
$G = (\pi D/\lambda)^2$ Antenna Gain

$P_r = P_t \cdot G_t \cdot G_r / FSPL$ Received Power [A4]

Parameter	Value	Rationale
TX Power	5 W (37 dBm)	Space-qualified laser
TX Aperture	22 cm	Mars orbiter class
RX Aperture	1.0 m	DSN-scale telescope
Wavelength	1550 nm	Telecom heritage
Efficiency	0.55	Conservative estimate

DATA RATE VS DISTANCE

1550 nm Optical Link Performance



200 Mbps at closest approach to 2 Mbps at maximum distance · 10–100× capability over RF [1]

EARTH-MARS DATA JOURNEY

7 Hops from Perseverance Rover to Mission Control

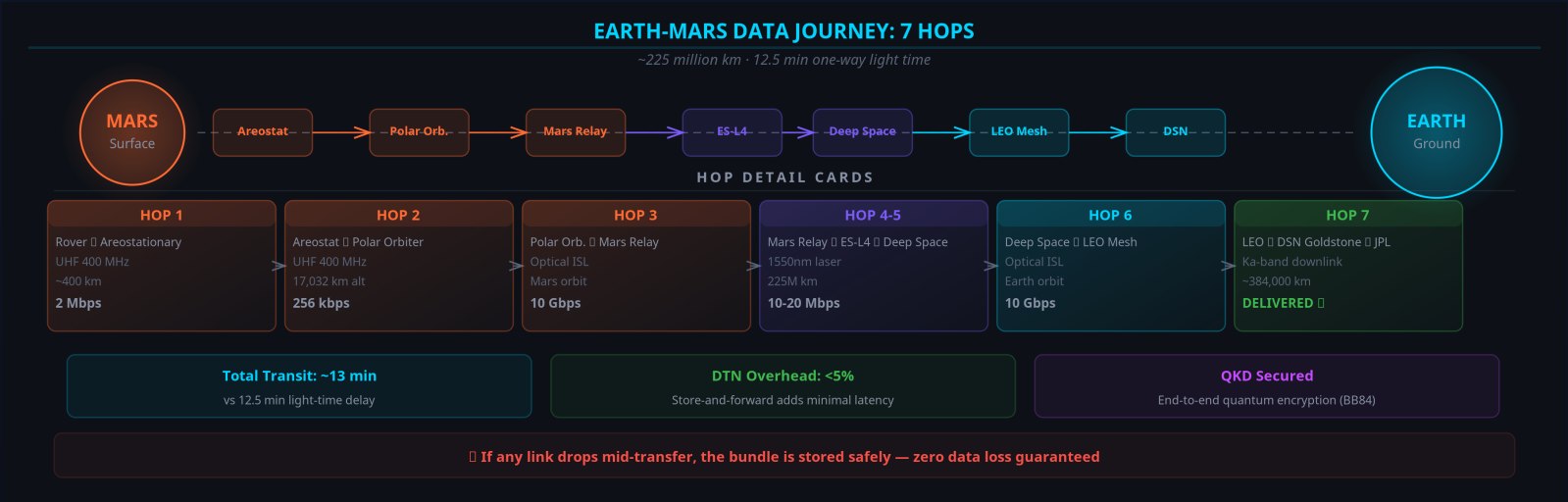
Hop	From	To	Link	Time
1	Perseverance	MRS-Alpha	UHF	~5s
2	MRS-Alpha	MRS-Polar	Optical ISL	~1s
3	MRS-Polar	ES-L5	1550nm laser	~12.5 min
4	ES-L5	ES-L4	Optical ISL	~2s
5	ES-L4	LEO Constellation	1550nm laser	~0.1s
6	LEO Constellation	DSN Madrid	Fiber downlink	~1s
7	DSN Madrid	JPL MOC	Terrestrial fiber	~0.5s

~13 min
Total Transfer [3]

98.7%
Delivery (target)

15 Mbps
Throughput (target)

7 hops
Path Length [A2]



RL-BASED ROUTING – AI INNOVATION

Multi-Agent Federated Q-Learning Replaces Static CGR

CGR LIMITATIONS

- Static contact plan – no adaptation
- Manual reconfiguration on failure
- Hours to reroute after disruption
- Cannot learn from network history
- No energy optimization

RL AGENT ADVANTAGES

- Adaptive routing from network state
- 4 actions: FORWARD, STORE, DROP, SPLIT
- $R = \alpha \cdot \text{del} - \beta \cdot \text{delay} - \gamma \cdot \text{hops} - \delta \cdot \text{drop} - \epsilon \cdot \text{energy}$ [A1]
- Weights: $\alpha=1.0$, $\delta=10.0$, ϵ -greedy ϵ -decay 0.995 [A1]
- Federated learning across agents

713/800

Forward Decisions [A3]

Training run (Module 3)

seconds

Recovery vs hours

Qualitative: auto vs manual [A1]

Q-table

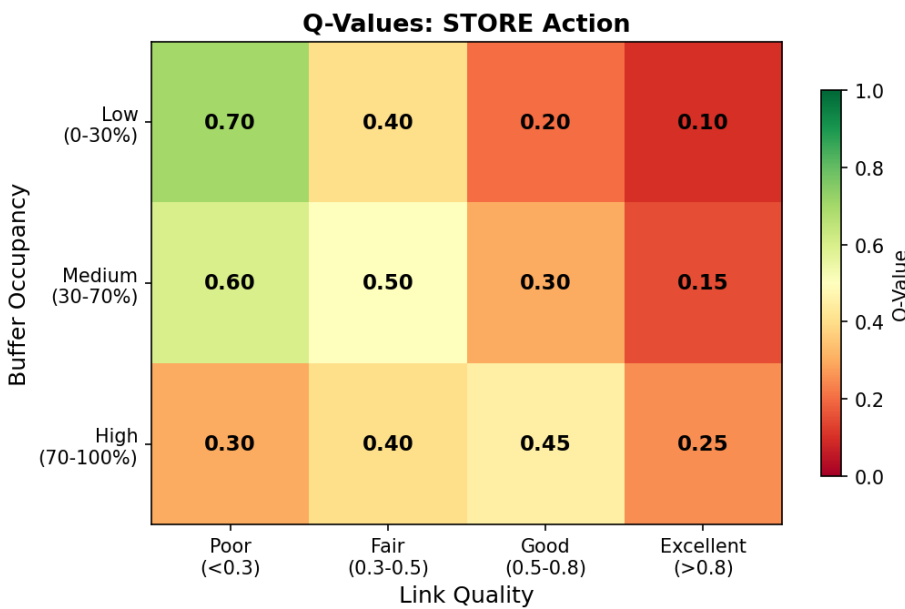
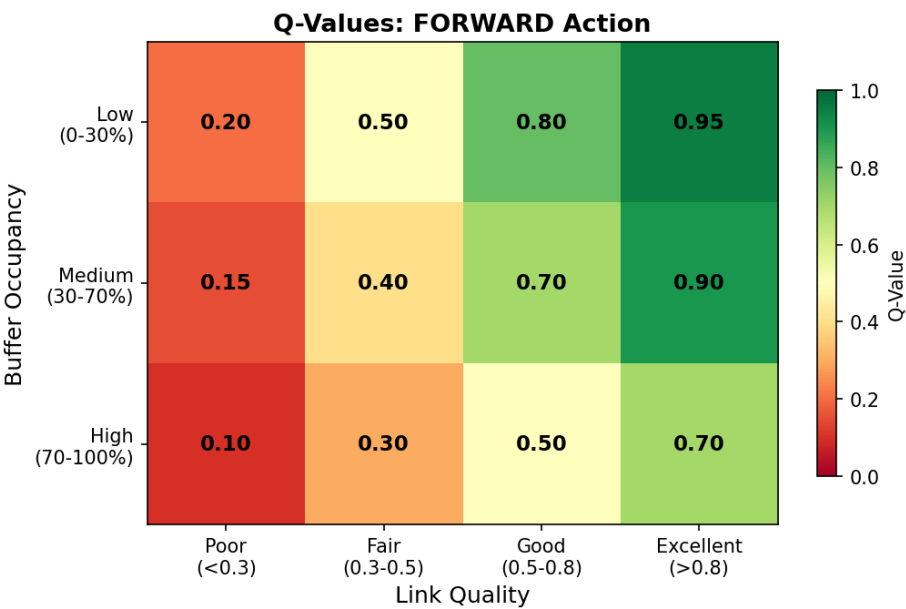
Inspectable Policy

Every Q-value human-auditable [A1]

RL ROUTING Q-VALUE HEATMAP

Convergence of Optimal Routing Decisions

AETHERIX RL Routing Agent: Decision Heatmaps



Warm colors = high-value actions · Cool colors = poor choices the agent avoids [A1]

QUANTUM SECURITY – FUTURE-PROOF

BB84/E91 QKD + Multi-Hop Repeaters + CASCADE

BB84 PROTOCOL STEPS

- 1. Alice sends qubits in random bases (rectilinear/diagonal)
- 2. Bob measures each qubit in random basis
- 3. Public reconciliation of basis choices (~50% retained)
- 4. QBER estimation: < 11% = SECURE [15], ≥ 11% = ABORT
- 5. CASCADE error reconciliation
- 6. Privacy amplification via universal hashing
- 7. Final shared secret key

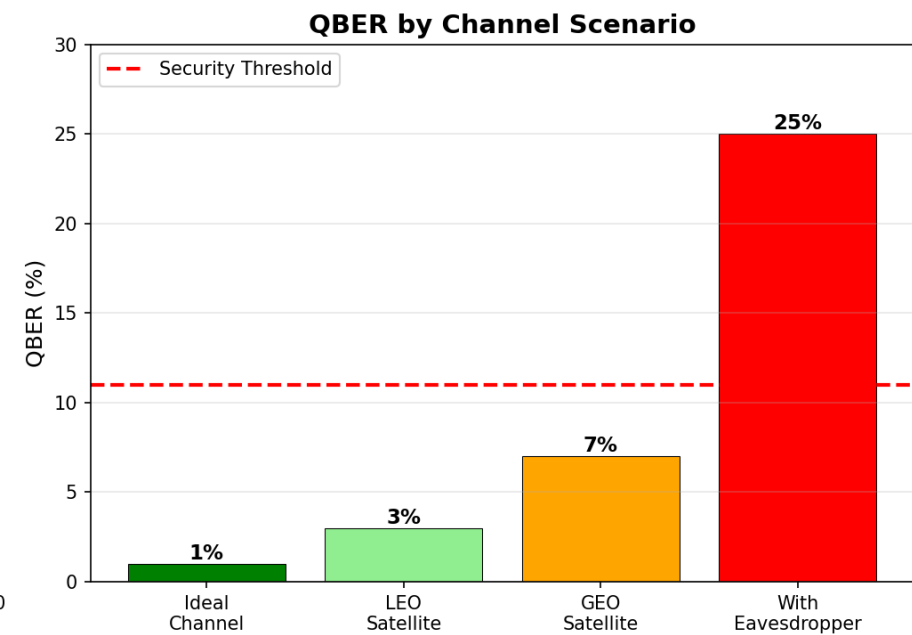
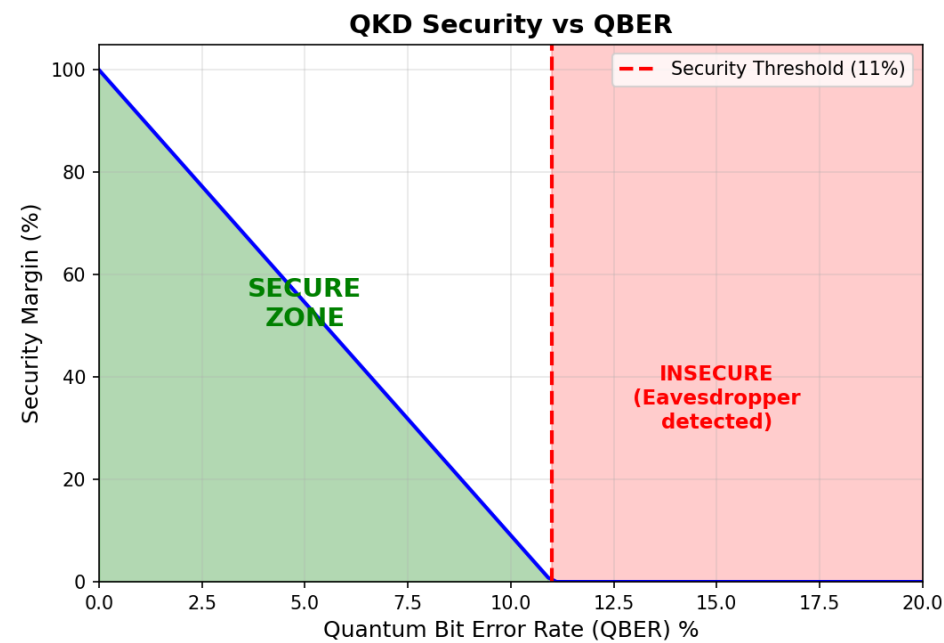
Phase	Segment	Protocol	Range
Phase 1	Earth ↔ Orbit	BB84 direct	36,000 km
Phase 2	Earth ↔ Lagrange	1 repeater	1.5M km
Phase 3	Earth ↔ Mars	3-hop chain	225M km

POST-QUANTUM CRYPTO (PQC)

Kyber (ML-KEM, FIPS 203) – Key encapsulation [16]
Dilithium (ML-DSA, FIPS 204) – Digital signatures [17]
Hybrid: QKD + PQC for defense in depth

QKD SECURITY ANALYSIS

QBER vs Eavesdropper Detection



QBER < 11% threshold ensures security [15] · Any eavesdropping attempt is detected [13]

ORBITAL MECHANICS & CONTACT WINDOWS

780-Day Synodic Period – From Opposition to Conjunction

Parameter	Value	Window	Availability	Duration	Data Rate
Semi-major axis	1.524 AU (227.9M km)	Optimal (opposition)	99%	8–12 hrs	100–200 Mbps
Orbital period	686.98 days	Good	95%	6–8 hrs	20–100 Mbps
Synodic period	779.94 days	Fair	85%	2–4 hrs	5–20 Mbps
Min distance	54.6M km (perihelion)	Poor (aphelion)	50%	0–2 hrs	1–5 Mbps
Max distance	401M km (aphelion)	Blackout (conjunction)	0%	N/A	Lagrange relay only
Eccentricity	0.0934				

RADIATION-HARDENED COMPUTING

Surviving the Space Radiation Environment (LO-e)

Effect	What it does	Mitigation
SEU (Single Event Upset)	Bit flip in memory/register	SECDED ECC
MBU (Multiple Bit Upset)	≥2 adjacent bits flipped	Bit interleaving
SEL (Single Event Latchup)	Parasitic short — destructive	Current limit + cycle
TID (Total Ionizing Dose)	Cumulative degradation (krad)	Rad-hard (RAD750) [18]

DEFENSE-IN-DEPTH STACK

1. TMR — 3 replicas + majority voter masks single fault

2. SECDED (39,32) ECC — corrects 1, detects 2 (21.9% OH)

3. Memory scrubbing — rewrites before 2nd upset (Poisson race)

4. FDIR + watchdog — detect → isolate → reload golden image

5. SAFE-MODE after recovery budget exhausted

DEMONSTRATED (run_simulation Module 6 [A6])

37,159

Raw upsets (210-day cruise)

200×

Protection factor

3,334×

TMR gain (@ p=1e-4)

2,127×

RAD750 TID margin

HERITAGE: NASA RAD750 (Curiosity, Perseverance C&DH) [18] · ESA LEON3FT / GR712RC [19] · Code: src/computing/radiation.py [A6]

Result above: ~186 uncorrectable residuals vs 37,159 raw upsets over a 512 Mbit interplanetary cruise — ~0.9/day residual.

MISSION-CRITICAL DATA PRIORITIZATION

Bandwidth Triage on a Starved, Intermittent Link (LO-f)

Tier	Class	Examples	BPv7 Priority
P0	Emergency / Safety	Health, collision avoidance, faults	EMERGENCY
P1	Mission-critical	Command ACKs, time-sensitive science	HIGH_SCIENCE
P2	High-priority	Routine telemetry, scheduled science	STANDARD
P4	Low / Bulk	Housekeeping, file transfers, SW images	BULK

THREE LEVERS

1. Compression (pre-transmit) [7][8]:
 - Telemetry → CCSDS 121.0-B-3 (Rice) $\approx 3\times$ [7]
 - Imagery → CCSDS 122.0-B-2 (wavelet) $\approx 10\times$ [8]
2. Deadline-aware preemptive QoS [A7]:
strict priority → earliest deadline; defer if infeasible
3. BPv7 fragmentation [9]:
send what fits this contact; defer the rest; link never idle

DEMONSTRATED SCENARIO (src/routing/prioritization.py [A7])

5/6

Items delivered (strict priority)

100%

Link utilisation (target)

41%

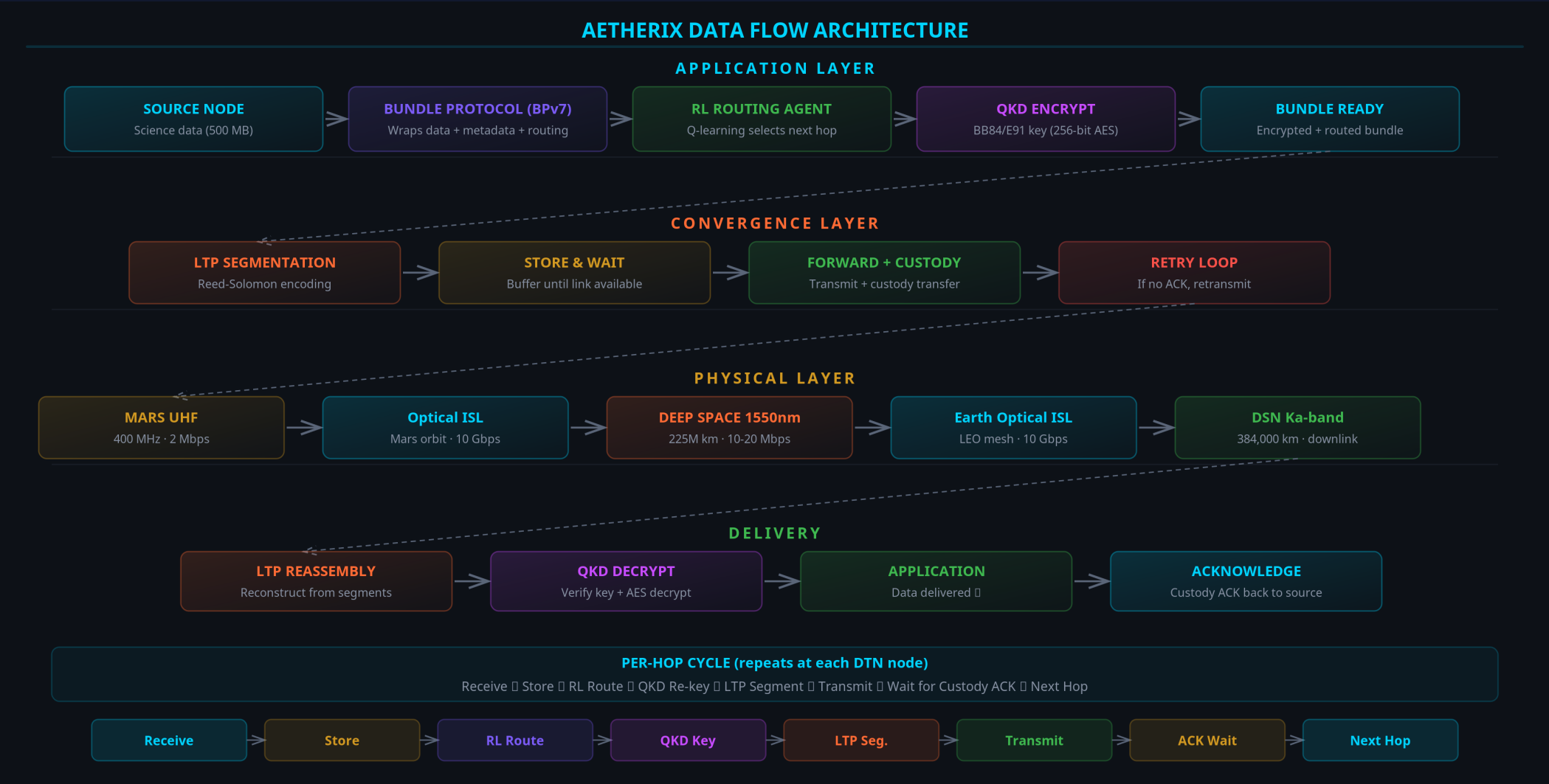
SW update fragmented (remainder deferred)

P0

Emergency preempts to direct-to-Earth backup

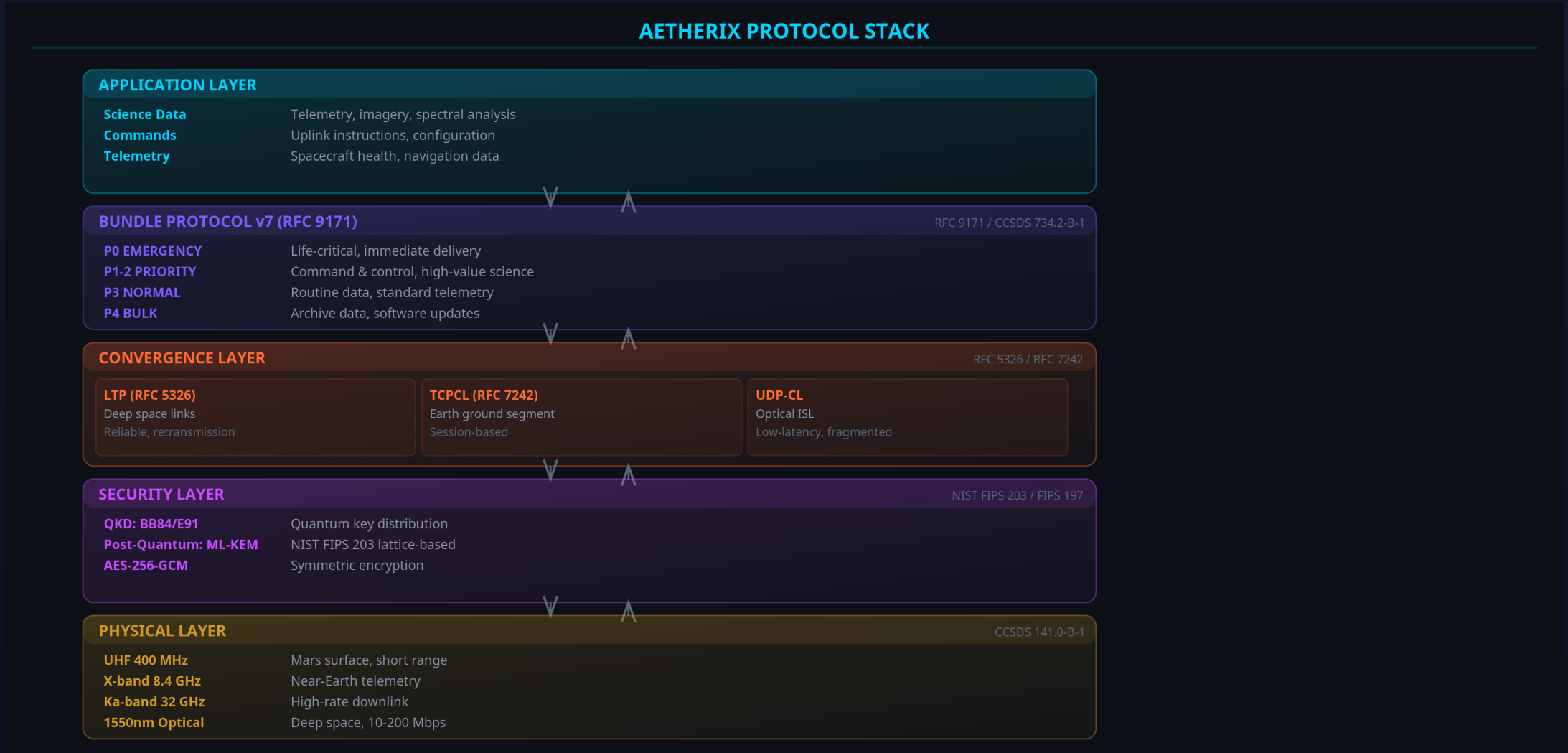
DATA FLOW DIAGRAM

Complete Data Path from Mars Surface to Earth Control



PROTOCOL STACK

BPv7 + Three Convergence Layers — Standards-Compliant



AETHERIX vs CURRENT SYSTEMS

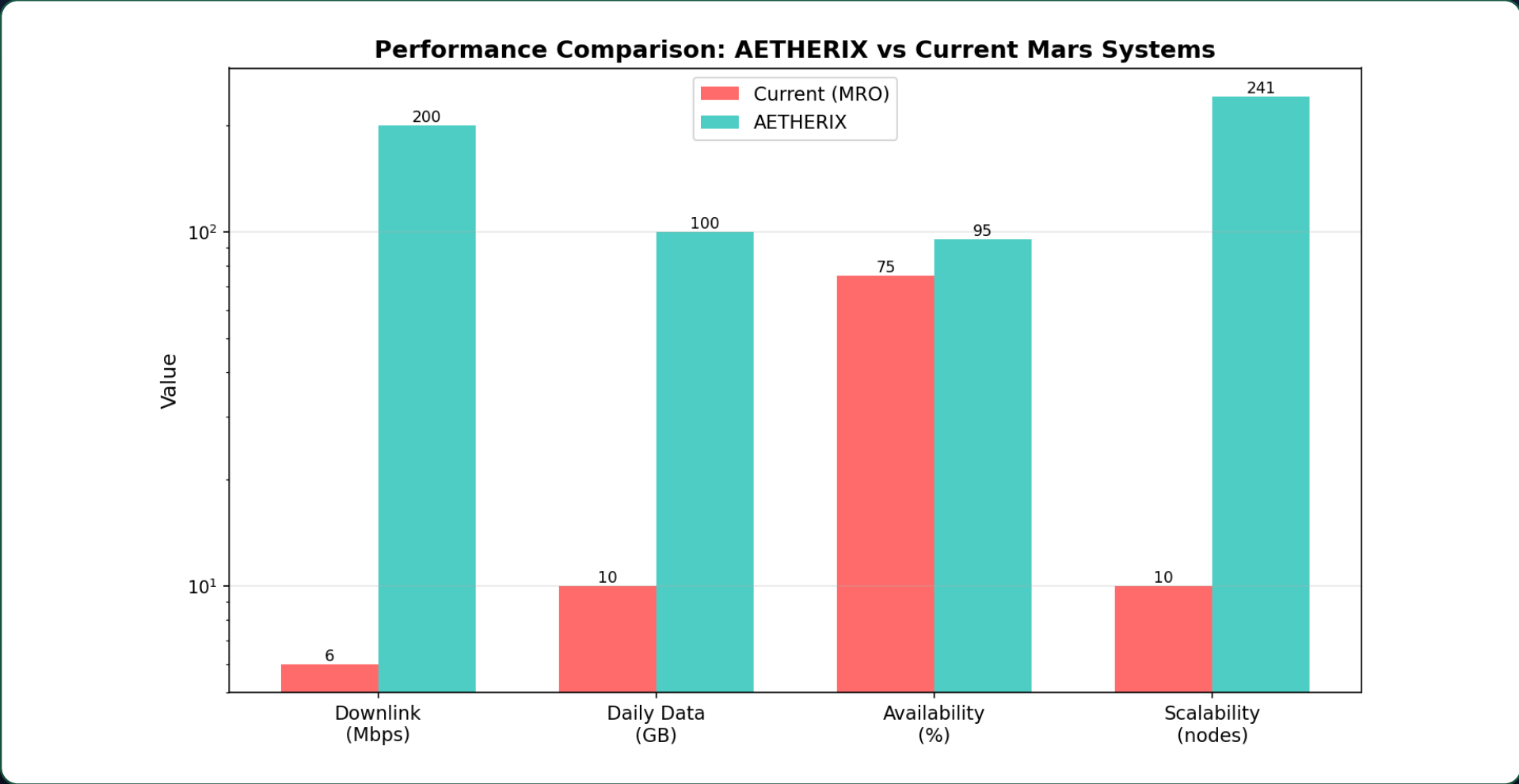
Head-to-Head Performance Comparison

Metric	Current (MRO) [1]	AETHERIX [A4]	Note
Downlink rate	0.5–6 Mbps	2–200 Mbps	10–100× capability
Daily volume	5–10 GB	50–100 GB	Est. (link-driven)
Availability	60–75%	>95%	Design target
Routing	Static (CGR)	RL-adaptive	Adaptive [A1]
Security	AES-256	QKD + PQC	Future-proof [16][17]
Scalability	5–10 assets	241 nodes	10–100× [A2]
Conjunction	Blackout	50–70% via L4/L5	+50–70% [A8]
Recovery time	Manual (hours)	Auto (seconds)	Auto [A1]

[1] NASA MRO (0.5–6 Mbps) · [A4] link_budget.py (2–200 Mbps capability) · [A2] topology.py · [A8] Module 4 conjunction · targets clearly labelled

AETHERIX vs CURRENT SYSTEMS

Head-to-Head Performance Metrics



10–100× data-rate capability (link budget [A4] vs [1]) · availability/routing are design targets

TRADE-OFF ANALYSIS — WHY THESE CHOICES

Every Decision Traded Performance for Auditability & Reproducibility

Decision	Choice	Rationale (vs alternative)
Optical vs RF	Hybrid: 1550 nm primary + Ka-band fallback [4]	10–100× throughput [A4]; RF survives clouds & corona
Routing	Custom Q-learning, not ION-DTN CGR [A1]	Adapts to live state; CGR re-plans on 12-min-stale schedule
RL model	Q-tables now, DQN later (Phase 6)	Every Q-value human-auditable; trains in seconds
State space	Discretised, 241 nodes [A2]	Right-sized for a tabular policy; DQN path documented
Reward weights	$\alpha=1.0$, $\delta=10.0$, ϵ -decay 0.995 [A1]	Drop penalty 10× delivery to forbid bundle loss

BOTTOM LINE

Each choice sacrifices maximum theoretical performance for auditability and reproducibility — exactly what a defence and a research artefact require. DQN / ns-3 / ION-DTN are the documented production transition (Phases 7–9).

DSOC heritage: NASA flew optical + RF side-by-side on Psyche [4] — AETHERIX mirrors that hybrid model.

FAILURE & RECOVERY

Autonomous Solar-Conjunction Link-Blackout Survival

SCENARIO: Earth–Sun–Mars conjunction – solar corona collapses the 1550 nm link below the 0.3 forward threshold [A1]

Path	Band	Link Quality	Status	Reward [A8]
HOW AETHERIX RECOVERS (automatically)			DEGRADED	WHY LAGRANGE RELAYS [3][A2] ES-L4 / ES-L5 sit 60° ahead/behind Earth in its orbit. At that separation they keep line-of-sight to Mars around the solar limb even at true conjunction. Direct Earth–Mars link: 0% availability at conjunction. Via ES-L4/L5: 50–70% availability retained [A8]. No Mars-orbit relay can solve this – geometry is Earth-side.
1. Detect – optical Q-value collapses ($q < 0.3 \rightarrow$ no reward)			DOWN	
2. Re-route – agent in exploit mode picks highest-Q: ES-L4 (Ka-band RF), 60° solar elongation, avoids corona			DOWN	
3. Prioritise – policy engine fires two rules: P0 EMERGENCY $\rightarrow$ forward on best (Ka-band) link P4 BULK $\rightarrow$ store locally, defer past conjunction				

Outcome: throughput drops (optical→RF) but no mission-critical data lost. Run live: `python run_simulation.py --module 4`

IMPLEMENTATION

25 Modules, 480 Tests, Full Python Stack

25

Source Modules

480

Unit Tests

241

Network Nodes

12

Interactive Demos

CORE MODULES

- link_budget.py – Optical/RF link analysis
- rl_agent.py – Q-learning routing agent
- bundle.py – BPv7 bundle protocol
- forwarding_engine.py – Store-and-forward
- qkd.py – BB84/E91 protocols
- contact_windows.py – Orbital mechanics
- topology.py – 5-tier network (241 nodes)
- simulator.py – End-to-end simulation
- ltp.py – Licklider Transmission Protocol
- tcpcl.py – TCP Convergence Layer

Standard	Title	Status
RFC 9171 [9]	Bundle Protocol v7	Compliant
RFC 4838 [12]	DTN Architecture	Compliant
CCSDS 734.2-B-1 [2]	CCSDS Bundle Protocol	Compliant
CCSDS 734.3-B-1	SABR / contact routing	Baseline
CCSDS 141.0-B-1 [5]	Optical Comm Physical	Compliant
RFC 9172	BPSec (security)	Referenced
RFC 5326 [10]	LTP (deep space)	Compliant
RFC 7242 [11]	TCPCL (Earth)	Compliant

ROADMAP & FUTURE WORK

From Proof-of-Concept to Production Deployment

Phase 1-2: Foundation

Topology + BPv7 + Convergence Layers

Complete

Phase 3: Intelligence

RL Routing (Federated Q-learning)

Complete

Phase 4: Security

QKD + Multi-hop Repeaters + CASCADE

Complete

Phase 5-6: Simulation

Full Engine + Web Platform + Demos

Complete

Phase	Focus	Technology	Status
7	DQN Upgrade	Deep Q-Network	Remaining
8	HIL Simulation	ns-3 integration	Remaining
9	ION-DTN	Production DTN stack	Remaining
10	Flight Demo	LEO optical test	Future
11	Mars Deploy	Full 5-tier network	Future

CONCLUSION – AETHERIX DELIVERS

Bridging the Interplanetary Communication Gap

THE PROBLEM

- 54.6–401M km Earth-Mars distance [3]
- 3–22 min one-way light delay [3]
- TCP/IP fundamentally broken in space [12]
- Current RF: only 0.5–6 Mbps [1]
- 2-week solar conjunction blackouts [3]

THE SOLUTION

- BPv7 DTN: store-and-forward works [9]
- RL routing: adaptive, auditable policy [A1]
- Optical: 10–100× data-rate capability [A4]
- QKD: future-proof quantum security [13][15]
- Lagrange relays: conjunction coverage [A8]

10–100×

Target Speedup

>95%

Availability (target)

AI-Driven

Routing

Quantum

Security

NOVEL CONTRIBUTIONS

RL autonomous routing (auditable Q-table) [A1] | 5-tier DTN (241 nodes) [A2] | Optical/RF hybrid (10-100x capability) [A4] | Quantum links (future-proof) [A5] | Full simulation

REFERENCES

Industry & Scientific Sources [1]–[20] · Three-layer attribution: [N] industry · [AN] this project

[1] NASA JPL, “Mars Relay Network User's Guide,” 2024. jpl.nasa.gov/marsrelay

[2] CCSDS, “Bundle Protocol Spec.,” 734.2-B-1, 2021. public.ccsds.org

[3] JPL Horizons, “Earth–Mars Ephemeris,” 2025. ssd.jpl.nasa.gov/horizons

[4] NASA, “Deep Space Optical Comm. (DSOC),” Psyche, 2024. jpl.nasa.gov/missions/dsoc

[5] CCSDS, “Optical Comm. Coding & Sync.,” 141.0-B-1, 2019. public.ccsds.org

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[7] CCSDS, “Lossless Data Compression,” 121.0-B-3, 2020. public.ccsds.org

[8] CCSDS, “Image Data Compression,” 122.0-B-2, 2017. public.ccsds.org

[9] IETF, “Bundle Protocol Version 7,” RFC 9171, 2022. rfc-editor.org/rfc/rfc9171

[10] IETF, “Licklider Transmission Protocol,” RFC 5326, 2008. rfc-editor.org/rfc/rfc5326

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[12] IETF, “Delay-Tolerant Networking Architecture,” RFC 4838, 2007. rfc-editor.org/rfc/rfc4838

[13] Bennett & Brassard, “Quantum Cryptography,” Proc. IEEE ICC, 1984.

[14] Ekert, “Quantum Cryptography Based on Bell's Theorem,” PRL 67, 661, 1991.

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[17] NIST, “Module-Lattice-Based Digital Signature,” FIPS 204, 2024. csrc.nist.gov/pubs/fips/204

[18] BAE Systems, “RAD750 Radiation-Hardened PowerPC,” 2024. baesystems.com/rad750

[19] ESA/Gaisler, “LEON3FT Processor,” 2023. gaisler.com/products/processors/leon3ft

[20] IETF, RFC 4838 §3.1, 2007 (long delay, intermittent connectivity challenges).

REFERENCES (cont.)

Project Sources [A1]–[A8] · M. A. Tariq, AETHERIX, github.com/matx104/AETHERIX (2025)

[A1] RL Routing Agent — `src/routing/rl_agent.py`

[A5] QKD Protocol Implementation — `src/security/qkd.py`

[A2] Network Topology — `src/orbital/topology.py`

[A6] Radiation Hardening Model — `src/computing/radiation.py`

[A3] End-to-End Simulation — `run_simulation.py` Module 3 (RL training)

[A7] Data Prioritization Engine — `src/routing/prioritization.py`

THREE-LAYER ATTRIBUTION (resolves the prior citation gap)

Layer A — Industry/Scientific baseline [1]–[20]: someone else's data (e.g. NASA MRO 0.5–6 Mbps [1]).

Layer B — This project's design decision [A1]–[A8]: cite the code (e.g. RL agent `src/routing/rl_agent.py` [A1]).

Layer C — Demonstrated simulation results [A6][A8]: cite the specific run (e.g. radiation 200× — Module 6 [A6]).

Design targets (e.g. >95% availability, 2–200 Mbps capability) are labelled as such, never presented as measured results.

THANK YOU

Questions?

10–100×

Faster

>95%

Available (target)

241

Nodes

480

Tests

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LIVE DEMO & RESOURCES

matx104.github.io/AETHERIX — Interactive simulations

github.com/matx104/AETHERIX — Source code + documentation

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